TOPIC 2 Systems of Linear Equations and Inequalities

Name _____ Date _____

Reminders
Linear combinations: make one variable's coefficients opposites, then add the equations.
Multiply every term, including the number on the right side.
Fractions: multiply by the LCD first. Variable alone? Substitution is faster.

1. Solve: $3x + y = 7$ and $2x + 2y = 6$.

Tip: Multiply the first equation by −2.

A. (1, 2)

B. (2, 3)

C. (2, 1)

2. Plan A: 3 classes and 2 snacks cost $35. Plan B: 2 classes and 4 snacks cost $30. What does one class and one snack cost?

Tip: Check your answer in BOTH plans.

F. class $10, snack $2.50

G. class $2.50, snack $10

H. class $11, snack $1

3. Solve: $4x + 3y = 10$ and $3x + 2y = 7$.

> **Tip:** Multiply both equations so the y-terms are 6y and −6y.

× 2: 8x + 6y = _____

× (−3): −9x − 6y = _____

Add: −x = _____, so x = _____, y = _____